[OLD] Biometrics authentication on iOS and Android

In v3.23, we introduce new feature to test biometrics authentication on iOS devices during manual session

In v3.24, we add support to test biometrics authentication in automation session

In v4.0, we add support to test biometrics authentication on Android devices (for manual, automation and mixed session)

- Precondition
- Sending biometrics authentication response in manual session
- Sending biometrics authentication response in automation session
- Additional Notes

Precondition

You need to alter your application to use Kobiton custom library.

Refer to:

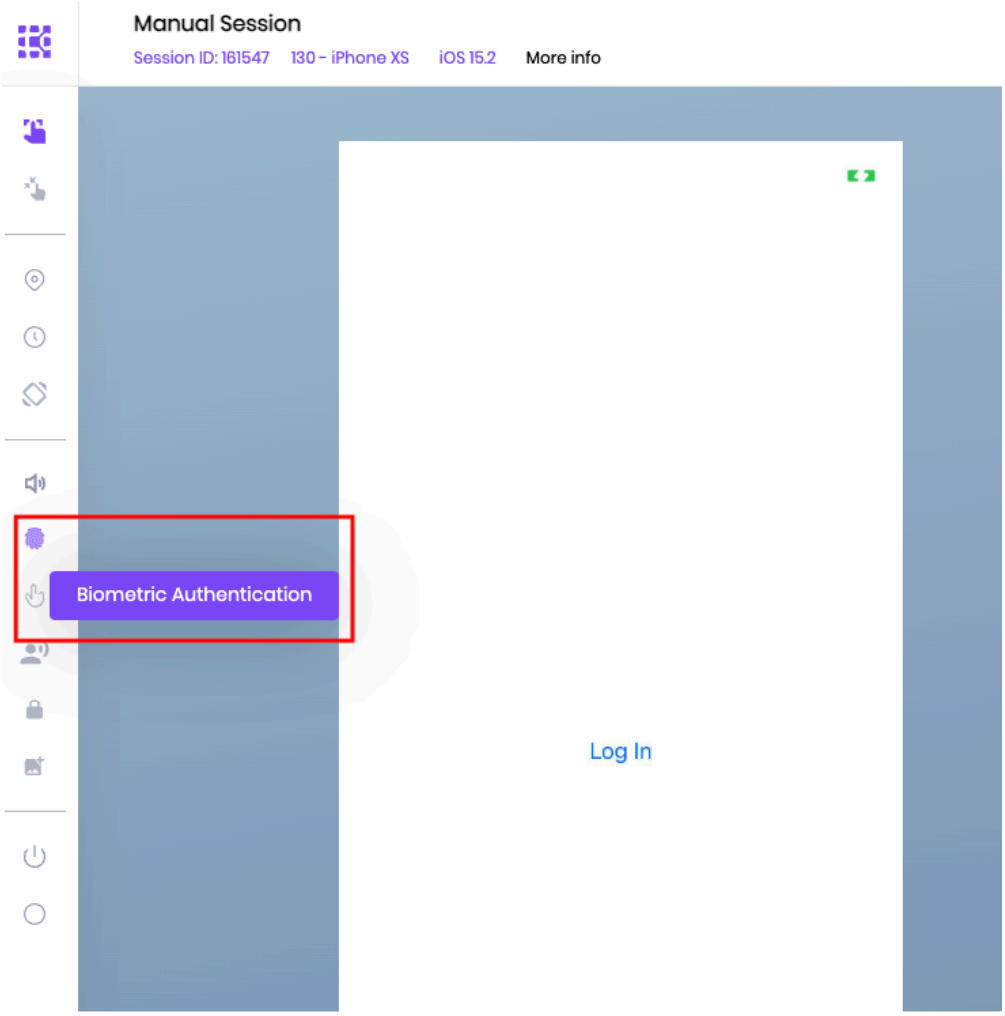
- iOS - Integration with user application for how to integrate with iOS app
- Android - Integration with user application for how to integrate with Android app

(@Dong Nhat Nguyen (Deactivated) please help to re-write this instruction)

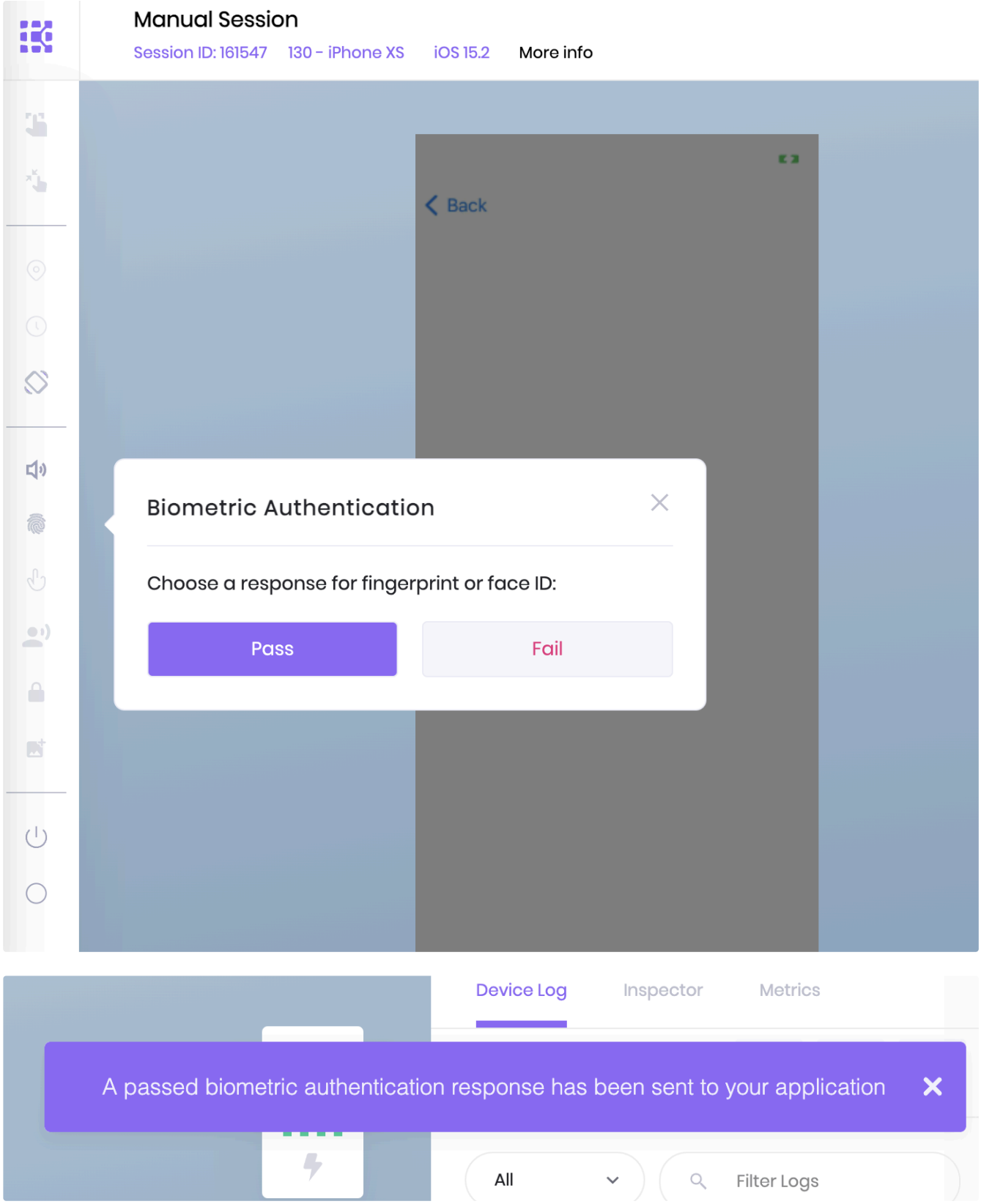
Once your application is ready, upload it to Kobiton App repo and use it in your test (manual/automation)

Sending biometrics authentication response in manual session

We added a new option on the left toolbar for biometric authentication:



When the application request for biometric authentication, Kobiton detects this request and prompt a dialog for you to choose a response. You can either choose Pass or Fail response.



There is a 20-second time-out, if you don't choose any response, after timed out, a Fail response will be sent to the device

While the device is waiting for your biometric authentication response, you won't be able to:

- Touch/Type/Swipe on the device

- use these features on left toolbar: set timezone, location, sensitive data, image injection, multi gesture

However, you will be able to use these features: adjust volume, click to return to Home screen, turn off/on screen, rotate

Sending biometrics authentication response in automation session

This is newly added in Kobiton version 3.24

Use the following commands to trigger a successful or failed authentication in response to a biometric prompt:

- `driver.execute('mobile:biometrics-authenticate', {result: 'passed'})` passes a successful authentication to the prompt.
- `driver.execute('mobile:biometrics-authenticate', {result: 'failed'})` passes a failed authentication to the prompt.

The following sample test script show how to use the command:

```
it('Run biometric auto', async () => {
  try{
    await driver.waitForElementByXPath('//XCUIElementTypeStaticText[@id="Log In"]').click()
    await driver.sleep(40000)
    await driver.execute('mobile:biometrics-authenticate', {result: 'passed'})
  }
  catch(err)
  {
    console.error(err)
    throw err
  }

})
```

Additional Notes

- This feature is only available for Latest Kobiton devices
- For iOS:
  - Our custom library to support biometric authentication requires minimum iOS 11. Hence, if your devices are in older iOS version, you won't be able to use this feature
  - This feature is only available for iPhone 5s and newer model
- For Android:
  - We support:

    Official API [https://developer.android.com/reference/android/hardware/biometrics/BiometricPrompt#authenticate(android.os.Cancel[...]cPrompt.AuthenticationCallback)](https://developer.android.com/reference/android/hardware/biometrics/BiometricPrompt#authenticate(android.os.Cancel[...]cPrompt.AuthenticationCallback))

Androidx Biometric 3rd library
https://androidx.de/androidx/biometric/BiometricPrompt.html#authenticate(androidx.biometric.BiometricPrompt.PromptInfo)

- Not support: more security with Crypto
  Official API
  https://developer.android.com/reference/android/hardware/biometrics/BiometricPrompt#authenticate(android.hardware.[...]cPrompt.AuthenticationCallback)
  Androidx Biometric 3rd library
  https://androidx.de/androidx/biometric/BiometricPrompt.html#authenticate(androidx.biometric.BiometricP[...]idx.biometric.BiometricPrompt.CryptoObject)
- We not yet verified this:
  https://developer.android.com/reference/android/hardware/fingerprint/FingerprintManager#authenticate(android.hardware.fingerprint.FingerprintManager.CryptoObject,%20android.os.CancellationSignal,%20int,%20android.hardware.fingerprint.FingerprintManager.AuthenticationCallback,%20android.os.Handler)

Demo video for this feature:

- manual (iOS): https://drive.google.com/file/d/1eH0th7SqaCIBEcF4E77tFf2YdmHJoiGB/view?usp=sharing Connect your Google account
- automation (iOS): https://drive.google.com/file/d/1lvosxYkgyL7ltTuA1KsipDEW51YcGW62/view?usp=sharing Connect your Google account
- manual (android): https://drive.google.com/file/d/17Ra96EiycO3HXdae-PUysZqGp_E05h8W/view?usp=share_link Connect your Google account
- automation (android): https://drive.google.com/file/d/1BXgFdrQXqymCp6CdJyhL40wbmurJ4AGv/view?usp=share_link Connect your Google account